

Low Sampling Rate Online Parameters Monitoring of DC–DC Converters for Predictive-Maintenance Using Biogeography-Based Optimization

B. X. Li and K. S. Low, *Senior Member, IEEE*

Abstract—Pulse width-modulated dc–dc converters are widely used in the renewable power systems. In practice, their reliability is of major concern. To identify the converter system, the Walsh–Hadamard transformation and recursive least square (RLS) methods are used in some previous works. However, the accuracy of the identified parameters is degraded under measurement noise. In this paper, a new approach for the full parameter estimation of a dc–dc buck converter is proposed. The new approach is realized using the biogeography-based optimization. Unlike the traditional RLS method, the proposed method is based on a state-space model with full-state observation. Therefore, the performance of parameter estimation under different measurement noise levels is improved. Both simulation and experimental results are presented to validate the effectiveness of the new estimation scheme.

Index Terms—Parameter estimation, online system monitoring, optimization methods, predictive maintenance.

I. INTRODUCTION

DC–DC converters are widely used in the power systems of electrical vehicles, avionics systems and renewable energy power plants. For these applications, the reliability of converters is a key concern. Previous industrial survey has shown that the electrolytic capacitors and switched devices caused 51% of the total failures in the converter circuits [1]. Therefore, they are mostly considered in the reliability analysis of the converter circuits [1]–[3]. Apart from the capacitor failure, the degradation of other converter components may also create unwanted voltage ripple and limit cycle oscillation at the converter output voltage. To improve the reliability of the power electronics, prognostic and health monitoring has been proposed in [4].

Various methods have been developed to detect the component wear out failures in the dc–dc converters [5]–[17]. The qualitative methods in [5] and [6] are based on the bond graph models. In these methods, the component failures are identified by applying a fault tree analysis. Although qualitative methods can be easily implemented with other part of the system to form a system-level fault detection, it can only be used to detect abrupt changes in the circuit by observing the failure symptoms.

To address this issue, quantitative methods are generally preferred. The state estimation method with Kalman filter is pro-

posed in [2]. The capacitor's equivalent series resistance (ESR) as well as the capacitance is estimated. The estimation results are then used for the capacitor's remaining life time prediction. Many of the reported methods [2], [3], [7]–[9] focus only on capacitor failure in the converter circuit. In [10], the full parameter estimation of the dc–dc converter is discussed. Methods reported in [2], [3], [7]–[10] require high sampling rate (usually 100 times higher than the switching frequency) due to the variations of the circuit topology in each switching cycle. According to [11], the high switching frequencies (e.g., more than 50 MHz) are expected in the future converter-on-a-chip design. With limited sampling capability, the performance of these methods will be affected due to the aliasing effect.

To realize the parameter estimation using a stand-alone system, lower sampling rate methods are explored in [12]–[17]. In these methods, the averaged models of the pulse width modulation (PWM) converters are applied. In [12]–[14], the impulse response of the buck converter is retrieved from the online measurements by applying the Walsh–Hadamard transformation or the discrete Fourier transformation. In [15], the zeros and poles of the open-loop buck converter system are directly estimated with RLS algorithm. The result is used to realize the online adaptive tuning of the PID controller. Although this method provides accurate and efficient computation for pole coefficients estimation and adaptive controller tuning, the experimental result shows coupling between the estimated zero coefficients due to the measurement noise. In [16], the modeling parameters in the boost converter are identified by measuring the step response. In [12]–[16], the system identification results cannot be directly applied to monitor the component degradations and in [17] only the series loss resistance is monitored.

In this paper, a low sampling rate and full parameter online estimation of a dc–dc converter approach is proposed. Using the averaged model, the online parameter monitoring is formulated as a multivariable optimization problem. The objective function is determined by the state estimation error based on the circuit measurements. To solve the optimization problem, the biogeography-based optimization (BBO) is proposed. BBO is a relatively new evolutionary algorithm and it can provide competitive results as compared to other optimization algorithms including genetic algorithm (GA) and particle swarm optimization (PSO) [18]. These computational methods aim to improve the solution of a multivariable optimization problem by sharing the information among the randomly initiated individuals iteratively. In GA, the unselected individuals with poor fitness are discarded directly. Due to the lack of search range diversity,

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The authors are with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798, Singapore (e-mail: libi0009@e.ntu.edu.sg; k.s.low@ieee.org).

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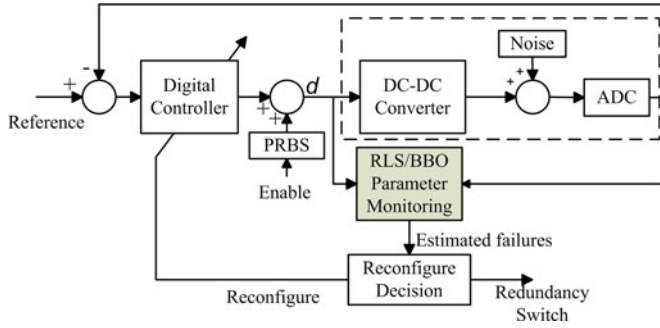


Fig. 1. Online parameter monitoring block diagram.

GA tends to converge prematurely. In PSO, the best feature in a particle is stored and perturbed with a random solution in each iteration disregard of its fitness. As a result, the dimension of the innovation in PSO is wide. The BBO provides an alternative selection scheme as compared with GA and PSO. A solution with poor fitness in BBO is given a lower chance to affect others. This is similar to GA. However, that solution in BBO is also given a higher chance to move to other solutions. The components in the circuit normally degrade slowly. Therefore, the population in the previous estimation could be added into the initial population in a new estimation to accelerate the convergence. For this reason, the BBO is preferred than other population-based optimization algorithms in this study.

The rest of this paper is organized as follows. In Section II, the proposed online parameter estimation methods are introduced. It includes the RLS and the BBO methods. The performance of the proposed method has been verified with simulation results in Section III. In Section IV, the experimental results are presented with a buck converter. Section V concludes this paper.

II. ONLINE PARAMETER MONITORING METHODS

Fig. 1 shows the overall block diagram of the proposed online parameter monitoring of a dc-dc converter. It consists of a PWM-controlled dc-dc converter, a digital controller, and a parameter estimation system. During the identification process, the pseudorandom binary sequence (PRBS) perturbation is applied. The voltage ripple constraint of the converter is assumed to be 5% of the nominal output voltage. The corresponding perturbation is therefore set to be $\pm 2.5\%$.

As shown in Fig. 1, the PWM duty cycle from the digital controller and the output measurements from analog-to-digital converter (ADC) are processed by the parameter estimation algorithm. The sampling time can be multiples of the PWM switching period. The ratio between them is termed as the sampling switch ratio (SSR) in the rest of this paper. This sampling scheme has been applied typically in the PWM converter system to allow the use of a low cost processor and reduce the power consumption [19], [20]. The parameter estimation results are

used to determine the component wear out failure. When a failure is detected, the controller will switch to the redundant converter. If the degradation is not serious, the digital controller can be tuned to adapt to the component degradations.

A. Average Model of DC-DC Buck Converter

The average model of the PWM converter is first proposed in [21]. It is also applied here to predict the dynamics in the buck converter circuit. Define the cycle-by-cycle averaged output voltage v_o and inductor current i_L as the state vector $\mathbf{x} = [v_o \ i_L]$. Moreover, let V_{in} and v_o represent the input voltage and cycle-by-cycle averaged output voltage, respectively. Then, the state-space model for the buck converter system can be derived as

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{u} \quad (1)$$

where

$$\mathbf{A} = \begin{bmatrix} -\frac{L + CRR_C}{CL(R + R_C)} & \frac{R(L - R_C R_S C)}{CL(R + R_C)} \\ -\frac{1}{L} & -\frac{R_S}{L} \end{bmatrix}$$

and

$$\mathbf{u} = \begin{bmatrix} \frac{V_{in} d R R_C}{L(R + R_C)} \\ \frac{V_{in} d}{L} \end{bmatrix}.$$

In (1), d is the input duty cycle. C, L, R, R_C, R_S are the capacitance, inductance, load resistance, capacitor ESR, and serial loss resistance. R_s includes three losses in the circuit namely the inductor ESR loss, the MOSFET loss, and the PCB trace loss. From (1), the transfer function of the buck converter is given in (2) as shown at the bottom of the page.

It is observed that delay is not considered in (2). This is because the sampling delay can be eliminated by synchronizing the ADC sampling with the PWM signal. Synchronization is typically adopted in the averaged model-based methods [12]–[17]. The sampling delay is compensated by the controller as a part of the synchronization procedure. Moreover, the load R and the input voltage V_{in} are assumed relatively constant for a short period of sampling time. While the dynamics of the load and input are expected to happen in practice, these dynamics will not have significant effect on the estimation accuracy. Based on the average model of buck converter in (1) and (2), two parameter estimation methods are considered in this paper and they are described in the sequel. Both methods are able to give accurate estimation of each component value.

$$\frac{V_o(s)}{D(s)} = \frac{V_{in}(RR_CCs + R)}{(R + R_s) + (L + RR_CCs + RR_SCs + R_S R_CCs)s + LC(R + R_C)s^2} \quad (2)$$

TABLE I
 θ VALUE ESTIMATION BASED ON RLS

Step 1: Initialization	
Set $P(1) = a * I$ and $\hat{\theta}(1) = \mathbf{0}$, where a is a large number, I is the identity matrix.	
for $k \geq 1$ repeat step 2 to step 4.	
Step 2: Update the P matrix	
	$P(k+1) = \frac{(I - \gamma(k)\varphi(k+1)')P(k)}{\lambda}$
Step 3: Calculate the innovation gain γ	
	$\gamma(k) = \frac{P(k)\varphi(k+1)}{\varphi(k+1)'P(k)\varphi(k+1) + \lambda}$
Step 4: Update the estimated $\hat{\theta}$	
	$\hat{\theta}(k+1) = \hat{\theta}(k) + \gamma(k)(y(k) - \hat{\theta}(k)\varphi(k))$

B. RLS Estimation and Parameter Mapping

In this section, the RLS estimation approach [22] is presented and it is used in this paper for benchmarking with the new BBO approach. Let k be the time index for each sampling. Define $\phi(k)$ as the data vector as follows:

$$\phi(k) = [v_o(k-1), v_o(k-2), d(k-1), d(k-2)] \quad (3)$$

where v_o and d are the converter output voltage and duty cycle, respectively. Let z^{-n} represents n step delay in time domain. A second-order discretized model can then be derived from (2) with a zero-order hold (ZOH) as

$$\frac{v_o}{d} = \frac{V_{in}(\theta_3 z^{-1} + \theta_4 z^{-2})}{1 + \theta_1 z^{-1} + \theta_2 z^{-2}} \quad (4)$$

where $\theta(k)^T = [\theta_1, \theta_2, \theta_3, \theta_4]$ is the parameter vector. From the regression model (4), the state prediction error is determined as

$$\varepsilon(k) = v_o(k) - \phi(k)\hat{\theta}(k). \quad (5)$$

The RLS algorithm solves for $\hat{\theta}(k)$ which can minimize $\sum_{k=1}^N \varepsilon(k)^2 \lambda^{N-k}$ through a series of recursive calculations as shown in Table I. Here, N is the number of the available measurements and λ is the forgetting factor. The forgetting factor is added so that the RLS estimations converge faster during the identification.

As shown in Table I, the algorithm continuously generates the estimation result $\hat{\theta}(k)$ to track the variations of the system online. In order to retrieve the component values from $\hat{\theta}(k)$, the parameter mapping in [23] is adopted. However, it gives two sets of solutions. The trivial solution is dependent on the SSR and the load resistance R . Therefore, the real solution can be identified by performing multiple estimations at different SSR and different load levels.

Although online monitoring can be achieved with the RLS method [15], the estimation result is sensitive to the measurement noise. Moreover, as there is dynamic dependence on the operating point of the current in the case of boost converter, a perturbation signal applied for the purpose of parameter estimation will cause the coefficients in the small signal model of the boost converter to change substantially. As a result, the accuracy of parameter estimation will be affected if RLS is applied to

a boost converter. To overcome these shortcomings, the BBO method is proposed.

C. BBO Parameter Estimation

BBO [18], [24] is often used to solve for multivariable optimization problems. It is inspired by the biogeographical immigration of the species among different habitats. When the habitat suitability index (HSI) is low, the solution will move to another habitat with a higher HSI. The improved performance of combining BBO with local search to determine the parameters of a dc-dc converter for predictive maintenance is explored in this paper.

Although the BBO objective function can be derived from the squared error of v_o in (4), it is beneficial to include more information in the optimization. Due to modeling approximation errors, the state-space estimation is more accurate than the transfer function estimation in the nonlinear systems such as boost converter. In the sequel, the squared sum of the state estimation error is used as the BBO objective function.

Given the initial states x_0 and d sequence in the state space (1), the subsequent states can be estimated as a function of the component values by solving an initial value problem. First, the eigenvectors and eigenvalues for matrix A are obtained as v_i and λ_i . The system state at time t is estimated as

$$\hat{x}(t) = \sum_{i=1}^2 c_i v_i e^{\lambda_i * t} + x_p \quad (6)$$

where c_i are constant coefficients of the eigenvectors and x_p is the particular solution of (1). In the cases, the imaginary eigenvalues are $\lambda = a \pm bi$ and the corresponding eigenvectors are $v = \begin{bmatrix} w_{11} \\ w_{12} \end{bmatrix} \pm \begin{bmatrix} w_{21} \\ w_{22} \end{bmatrix} i$, where $\begin{bmatrix} w_{11} \\ w_{12} \end{bmatrix} = w_1$, and

$$\begin{bmatrix} w_{21} \\ w_{22} \end{bmatrix} = w_2. \text{ The state estimations are determined as}$$

$$\hat{x}(t) = c_1 e^{at} (w_1 \cos(bt) - w_2 \sin(bt)) + c_2 e^{at} (w_1 \sin(bt) + w_2 \cos(bt)) + x_p. \quad (7)$$

c_i and x_p in (6) and (7) can be obtained from the initial condition x_0 as follows:

$$x_p = -A^{-1}u$$

$$c = Q^{-1}(x_0 - x_p) \quad (8)$$

where $Q = [v_1 \ v_2]$ and $[w_1 \ w_2]$, respectively, in (6) and (7). The objective function of the BBO algorithm is defined as the total squared state estimation error

$$J = \sum_{k=1}^N (x(k) - \hat{x}(k))^2 \quad (9)$$

where the estimated state $\hat{x}(k)$ is obtained from (6) and (7) by substitute t with the sampling period T and x_0 with the last step measurement. A total number of N samples are included in the objective function (9). The value of N is a design parameter and its selection will be discussed later. It is observed that the BBO method solves a minimum squared error problem. While

the RLS estimation can only reach the minimum squared error solution with the forget factor $\lambda = 1$ in the steady state. The selection of the forget factor λ in the RLS method is a tradeoff between the noise rejection and the convergence rate [22]. To reduce the total sampling and perturbation time, it is typical to choose $\lambda = 0.98$ [15]. Therefore, the RLS method cannot perform as good as the BBO method especially when the noise level is high. The steps for the proposed algorithm are given in Table II.

Most of the BBO parameters such as FE_{max} , N_{ls} , n_{ls} , and N_{ri} values are determined using nominal values as in other research works [24], [25]. The population size n_{pop} is usually set as ten times of the problem's dimension so that the search space is explored more often. The studies in [26] show that BBO is relatively robust to different mutation schemes and the mutation rate is typically selected as 0.01. The expected threshold J_{th} in Table II can be estimated from the noise level in the measurement by the definition of the deviation as

$$J_{th} = (1 + m) N \sigma^2 \quad (10)$$

where σ is the standard deviation of the measurement. For a reasonably large sample size N (e.g., $N \geq 50$) the margin $m = 0.3$ is adopted to guarantee the convergence. The computation cost of RLS parameter estimation has been discussed in [15] in the worst case scenario. The parameter mapping process requires only a few more elementary operations as shown in [23]. Comparatively, the computational load for BBO method is analyzed as well and the results are shown in Table III.

In each BBO iteration, $2n_{pop} + n_{ls}$ **MaxIter** function evaluations, $n_{pop}D$ habitat modifications, and two round of sorting with n_{pop} elements are involved. In each function evaluation, the number of addition and multiplication operation is given by $2N + A_{eig}$ and $2N + M_{eig}$, where A_{eig} and M_{eig} are the number of operations required to obtain the eigenvalues and eigenvectors from the BBO generated circuit parameters.

III. SIMULATION RESULTS

To verify the proposed online parameter monitoring methods discussed in Section II, both simulation and experimental studies on a buck converter have been conducted. The objective is to monitor the C , L , R_c , R_s values in the buck converter while it is operating in normal application. For the simulation, the PWM switching frequency is set at 20 kHz. Moreover, Gaussian white noise is added to the measured data with 2 mV deviation. This is to emulate the quantization error of the ADC converter with 12-bit resolution. Table IV shows the circuit parameter settings used in the study. In this paper, the nominal duty cycle is 33% and the perturbed duty cycle signal is set as $\pm 0.8\%$.

A. Simulation Result With RLS

The simulation result for the RLS parameter estimation is shown in Fig. 2. The forgetting factor λ in the RLS algorithm [15] is set to 0.98 to reject the output noise.

The simulation starts at time $t = 0$ s. At time $t = 0.02$ s, the PRBS perturbation is added into the duty cycle. The RLS algorithm then starts to estimate the transfer function coefficients.

TABLE II
PROPOSED PARAMETER ESTIMATION BASED ON BBO

Step 1: Initialization

Generate a population of n_{pop} individuals $\chi = [\hat{C}, \hat{L}, \hat{R}_c, \hat{R}_s]$. Each of χ contains a random estimation of the parameters within the given boundary.

Assign suitable values to the following BBO constants:

Problem dimension $D = 4$,
Population size $n_{pop} = 10D$,
Maximum function evaluation $FE_{max} = 150000$,
Threshold of objective function value J_{th} ,
Mutation rate $\xi = 0.01$,
Local search criteria $N_{ls} = 10$,
Number of solutions in local search $n_{ls} = 2$,
Re-initialization criteria $N_{ri} = 10$.

Set $counter = 0$.

While the number of function evaluations $FE \leq FE_{max}$ or $J > J_{th}$ repeat step 2 to step 7 for each generation.

Step 2: $counter++$

Step 3: Check the habitat suitability index HSI for each individual

The HSI for the k^{th} individual χ_k is defined as:

$$HSI(\chi_k) = \frac{1}{J_k},$$

where J_k is obtained from (9). The emigration rate μ_k and immigration rate ν_k are defined as:

$$\mu_k = \text{rank}(HSI(\chi_k)) / (n_{pop} + 1),$$

$$\nu_k = 1 - \mu_k,$$

where the function $\text{rank}(\cdot)$ gives the rank of the individual's HSI value in an increasing order. The best individual with the highest HSI is stored as χ_{best} .

Step 4: Habitat modification and mutation

for $k = 1; k \leq n_{pop}; k++$

for $i = 1; i \leq 4; i++$

if $\text{rand}_1 > \mu_k$

Select another individual l with Roulette wheel selection method so that the chance of each individual being selected is proportional to its immigration rate ν . Update the k^{th} individual with the step size proportional to ν_l .

$$\chi_k(i) = \chi_k(i) + \nu_l(\chi_l(i) - \chi_k(i))$$

if $\text{rand}_2 > \xi$

Reinitialize $\chi_k(i)$ with random number within the boundary

end

end

rand_1 and rand_2 are generated randomly in the range $[0, 1]$.

Step 5: Elite ranking

Rank the HSI again, replace the worst individual with previous χ_{best}

Calculate $\Delta \chi_{best}$ and update χ_{best} .

Update FE

Step 6: Local search with **fmincon** function in MATLAB Optimization Toolbox. The default parameter tuning of this function is used. E.g. the maximum iterations **MaxIter** = 1000.

if $\Delta \chi_{best} < N_{ls} * J_{th}$

Apply gradient decent search. The initial guess to be the n_{ls} best solutions in the population, replace the solutions with the local searched result.

Step 7: Re-initialization

if $counter \% N_{ri} == 0$

Add n_{pop} randomly generated individuals into the population.

Evaluate their HSI for each individual with (9)

Select the best n_{pop} from the population of $2n_{pop}$ individuals as the new population.

Update FE

TABLE III
COMPUTATIONAL COMPLEXITY OF RLS AND BBO METHODS IN EACH
ITERATION IN THE WORST CASE

Method	Multiplications	Additions	Others
RLS	$14 + M_n + M_{\text{map}}$	$14 + A_n + A_{\text{map}}$	Elementary functions
BBO	$(2n_{\text{pop}} + n_{ls} \text{MaxIter}) \cdot (2N + M_{\text{eig}}) + n_{\text{pop}} D$	$(2n_{\text{pop}} + n_{ls} \text{MaxIter}) \cdot (2N + A_{\text{eig}}) + 2n_{\text{pop}} D$	Sorting; Roulette wheel selections

TABLE IV
CIRCUIT PARAMETERS OF BUCK CONVERTER

C (mF)	L (mH)	R_c (Ω)	R_s (Ω)	V_{in} (V)	V_{ref} (V)	R (Ω)
0.33	0.22	0.025	0.075	10	3.3	1

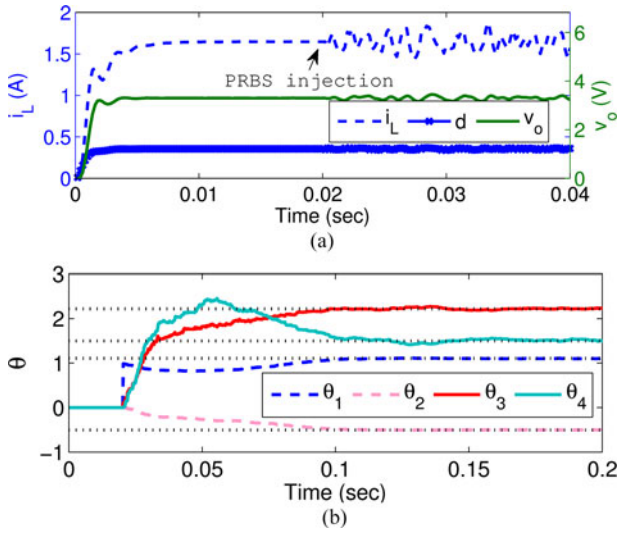


Fig. 2. (a) v_o , i_L , and d waveforms. (b) Estimation result and the theoretical values (dotted line) of the discrete model parameter θ .

The output voltage and inductor current are sampled at 5 kHz and the corresponding SSR is 4 for this test. The simulated waveforms of v_o , i_L and d sequences are illustrated in Fig. 2.

From Fig. 2(a), it is observed that the output voltage reached the reference voltage of 3.3 V after 0.01 s. The PRBS is enabled at 0.02 s. Small ripples within acceptable range ($\pm 5\%$) is observed at the output due to the perturbation. Fig. 2(b) shows the parameters estimation converged to the theoretical value after 0.1 s. The result can be mapped into the component values and they are shown in Fig. 3.

In Fig. 3, the estimation error is initially large but the results converged to the simulation settings after 0.1 s. The normalized mean square error for capacitance and ESR estimation are 1.8% and 5.4% for the estimation results during the time interval 0.2–0.3 s. This result is in agreement with the simulation results achieved in [2] and [7]. However, it is achieved at a much lower sampling rate of 5 kHz instead of 80 kHz in [2] and 2 MHz in [7].

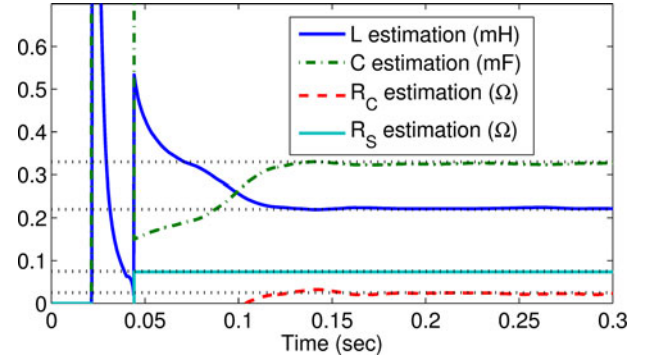


Fig. 3. Estimation result for buck converter circuit component and model parameter (dotted line).

TABLE V
BBO SEARCH RANGE FOR BUCK CONVERTER COMPONENT

	C (mF)	L (mH)	R_c (Ω)	R_s (Ω)
Model	0.33	0.22	0.025	0.075
BBO range	$[\varepsilon, 10\,000]$	$[\varepsilon, 10\,000]$	$[\varepsilon, 1]$	$[\varepsilon, 1]$

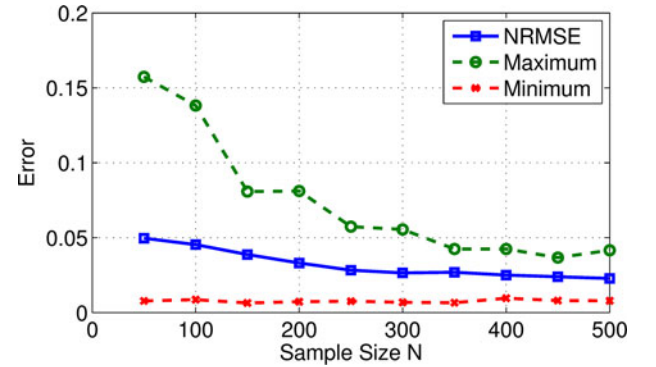


Fig. 4. BBO estimation accuracy at different sample sizes.

B. Simulation Result With BBO

The objective function of the BBO algorithm (10) is constructed with the same set of data as in RLS. The search range of BBO for the four parameters in the buck converter is defined to cover all possible values. Their values are listed in Table V. To investigate the suitable sample size for the construction of the BBO objective function, the sample size N in (10) is varied from 50 to 500 with a step size of 50 in simulation.

Fig. 4 shows the parameter estimation error at different sample sizes. The averaged normalized root mean square error (NRMSE) of the estimated $\hat{C}$, $\hat{L}$, $\hat{R}_c$, $\hat{R}_s$ are evaluated as follows:

$$\text{NRMSE} = \frac{1}{4} (E_{\hat{C}} + E_{\hat{L}} + E_{\hat{R}_c} + E_{\hat{R}_s}) \quad (11)$$

where $E_{\hat{X}} = \frac{1}{K \cdot X} \sqrt{\sum_{i=1}^K (X - \hat{X}_i)^2}$. In (11), X can be any one of the parameters C , L , R_c , R_s . For each sample size, the BBO

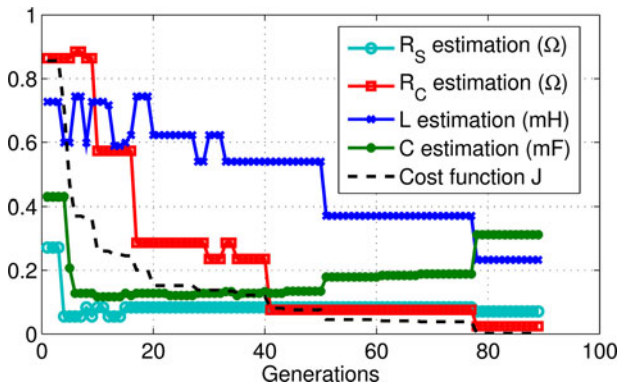


Fig. 5. BBO estimation error and evolution.

algorithm is run for $K = 100$ times with different measured data to form the objective function (9). From Fig. 4, it is observed that the estimation error decreases as more samples are used in the objective function. At the same time, the difference between the maximum and minimum estimation error is reduced. However, large sample size requires more computational power. After tradeoff, a sample size of $N = 300$ is selected for all the subsequent BBO estimations in this paper.

Fig. 5 illustrates the evolution process in the BBO estimation. The local search feature in the BBO algorithm allows all the four parameters to be updated at the same time. The results show that all the four parameters of the buck converter are estimated close to their settings. The final estimation is a stabilized solution because the cost function J converged to about 0.0012 which is expected according to (10) at the noise deviation of 2 mV.

C. Comparison of RLS and BBO Algorithm

To compare the performance of RLS and BBO algorithms, similar tests with different parameter settings have been conducted. In the first set of test, the loading scenario has been varied from 30% to 130% of the nominal load of 10.89 W. The v_o, i_L and d for BBO estimation and the RLS results are all collected in the same time interval [0.4 s, 0.5 s]. The BBO is repeated for $K = 10$ times with different samples in this interval. These ten estimations are then used for the NRMSE evaluation for BBO method at each point. The noise deviation for the simulation is set to 2 mV and the result is shown in Fig. 6

Fig. 6 shows that the accuracy of BBO is better than the RLS method. It is also observed that the performances of both algorithms are improved by increasing the SSR moderately. When the SSR is 1, which is the case in [15], the parameter estimation error is much greater than 30% due to the measurement noise.

To study the effectiveness of the proposed approach under noisy environment, the noise variation of the output voltage and inductor current measurements are increased in step of 30 mV² (mA²) starting from $\sigma^2 = 1$. The result is shown in Fig. 7.

Fig. 7 shows that the performance for both RLS and BBO estimation has improved with larger SSR under noisy environment. This is expected since the correlation between the duty

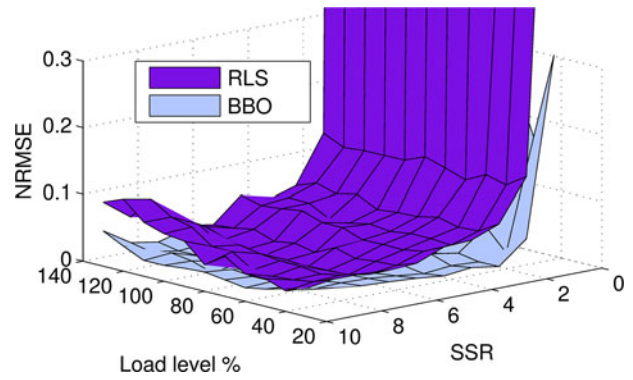


Fig. 6. Estimation accuracy under load variation.

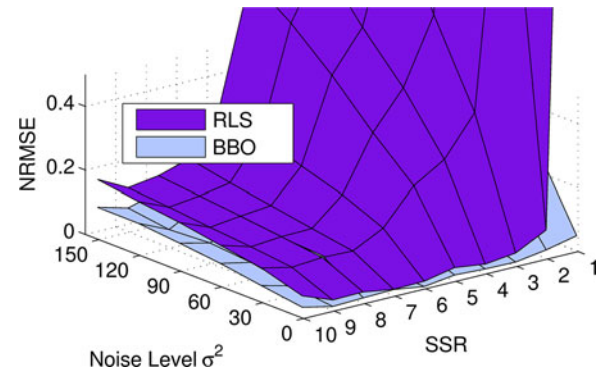
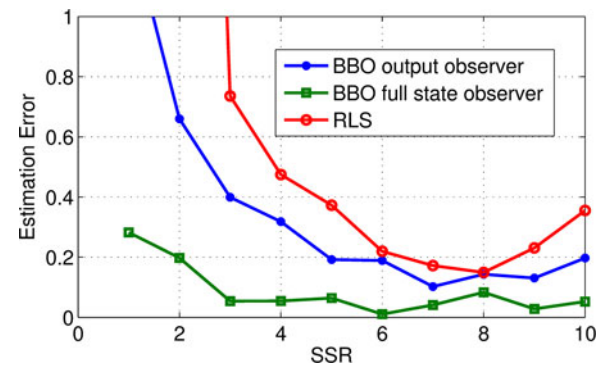


Fig. 7. Estimation accuracy under different measurement noise levels.

Fig. 8. Comparison of RLS method and BBO method with only the output data with $K = 10$ in (11).

cycle and the output voltage increased when the duty cycle is hold for a longer period.

To investigate the cause of different performance in RLS and BBO parameter estimations, the BBO objective function is derived from (4) with single v_o measurement. The new BBO algorithm is run under the noise variance of 31 mV² and the SSR is varied from 1 to 10. The result is depicted in Fig. 8 together with the result obtained for BBO and RLS in the noise level test.

It is observed in Fig. 8 that the BBO result is more accurate than the RLS method regardless of the number of measured

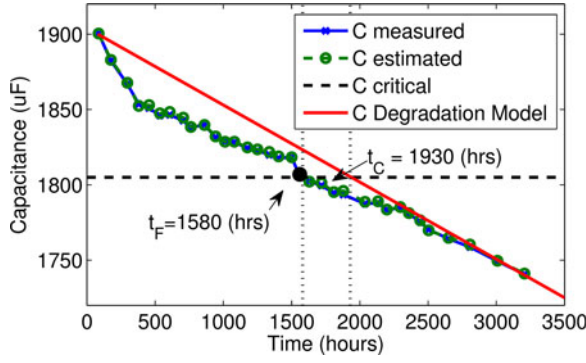


Fig. 9. Online monitoring of capacitor degradation detection.

states. However, the BBO algorithm has better performance with full-state estimation

D. Capacitor Ageing Simulation and Lifetime Estimation

The degradation of a capacitor causes its capacitance to reduce linearly with respect to time [27], [28]. In some applications with pulsed charging and discharging cycles, the capacitance loss may increase unexpectedly at certain pulses [29]. In the capacitor endurance test, 5% drop of the capacitance is usually considered as failure. In this case, the lifetime can be much shorten by the sudden capacitance loss. To estimate the lifetime correctly, the degradation model is updated after online parameter estimation with BBO. The degradation model of capacitor is given by

$$C(t) = C_0 + \xi t. \quad (12)$$

Without sudden loss, the capacitor lifetime is estimated according to the degradation model (12) as

$$\hat{t}_C = \frac{\Delta C}{\xi} = \frac{0.05 C_0}{\bar{\xi}}. \quad (13)$$

The degradation model parameters are extracted from the interpolated data of the thermal stress test result in [28]. Through offline measurements, the average degradation rate of the 15 capacitors at 105 °C is determined as $\bar{\xi} = 0.05 \mu\text{F/h}$. This feature is used to estimate the life time for each capacitor when online parameter monitoring is not available.

With online parameter monitoring, the capacitance value at each time is estimated as shown in Fig. 9. The degradation data for capacitor 13 in [28] are chosen as an example and used for simulation. The capacitor's ESR and other parameters are assumed to be remained constant as in Table III in the simulation.

Fig. 9 shows the measured capacitance and the estimated capacitance in the simulation. Without online parameter monitoring, the lifetime of the capacitor is estimated as $\hat{t}_C = 1930$ h. However, the 5% limit is exceeded at time $t_F = 1580$ h. The proposed online monitoring method is able to identify the capacitance at each time instant accurately and therefore can determine the end of life time of the capacitor correctly.

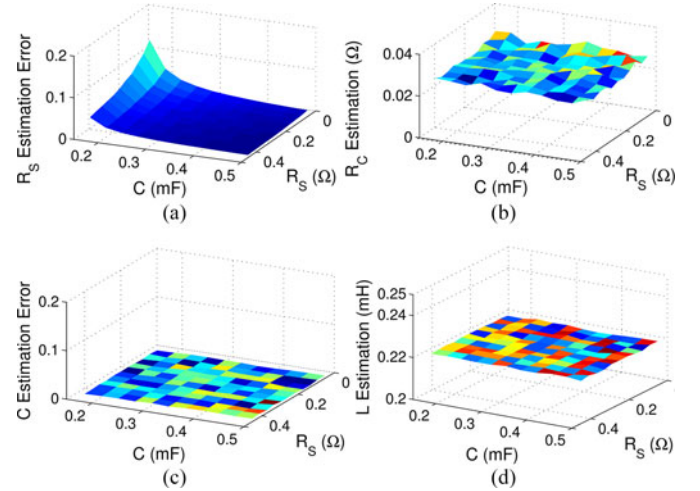


Fig. 10. (a) R_S error under parameter C, R_S variation. (b) R_C estimated under parameter C, R_S variation. (c) C error under parameter C, R_S variation. (d) L estimated under parameter C, R_S variation.

E. Monitoring of Multiple Parameter Variations

The performance of a dc–dc buck converter is dependent on the normal operation of each component in the circuit. When there are more than one component degrade, the proposed method is also able to detect and fully isolate them. The simulation is repeated with different C and R_S parameter settings. The capacitance is varied in the range of $\pm 50\%$ with a step of 10%. At the same time, R_S is varied from 0.04 to 0.44 Ω in step of 0.04 Ω . The BBO estimation result is depicted in Fig. 10. The normalized estimation error of the varied components and the estimated value of the normal components are evaluated in this test.

Fig. 10(a) shows that when C is large, the $\hat{R}_S$ estimation accuracy is independent from the variations in capacitance. It is observed that the estimation error increased to 10% with 50% capacitance loss due to the model inaccuracies. This will never happen in practice as the component will be replaced once serious degradation is found. Fig. 10(b)–(d) shows that the variations in the capacitance and inductor ESR will not affect the estimation of the other three components as the surfaces of the estimation results are almost flat.

F. Performance With Nonresistive Loading

The previous simulations are based on the assumption that the converter is connected with a pure resistive load. In actual circuit, the load might be passive or active. In the active load scenario, the output current and voltage relation for the supplying converter is depending on the load converter's switching states. In this case, the averaged system model is no longer valid. In the passive load scenario, the load capacitor must be connected in parallel with the output capacitor of the converter. The equivalent output capacitance is the sum of them. The following simulation aims to investigate the effect of the load inductance on the parameter estimation accuracy.

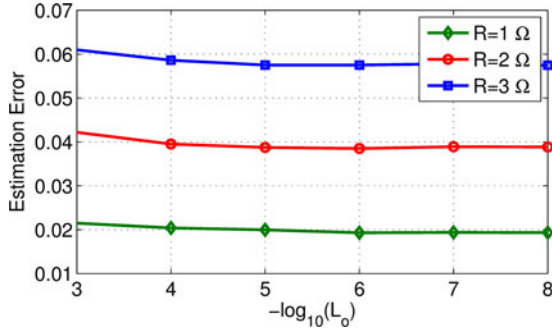
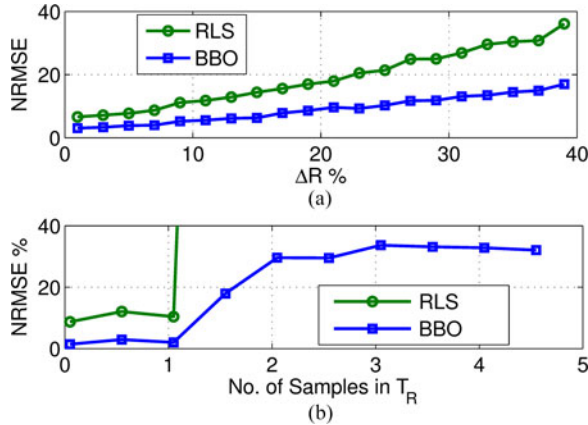


Fig. 11. Estimation accuracy under load variation.

Fig. 12. (a) Sensitivity tests at different ΔR with $T_R = T$. (b) Sensitivity tests at different T_R with $\Delta R = 0.05R_0$.

In this simulation, the load inductance L_o varying between 10^{-5} mH to 1 mH is connected in series with the load resistance R . The noise deviation is set to 1 mV at voltage measurement and 1 mA at the current measurement. The other simulation parameters are shown in Table III. The averaged estimation error is calculated with (11) and depicted in Fig. 11, which shows that the estimation accuracy is not affected by the load inductance.

G. Performance With Load Perturbation

The study of the sensitivity of BBO and RLS has been carried out through simulation with small perturbation on the load. The load profile is a square waveform with the original load R_0 and the perturbation ΔR . The period of the square waveform is T_R . The noise deviation is set to 1 mV and 1 mA so that the difference in the performance is mostly caused by the sensitivity of the different algorithms rather than the measurement noise. The switching frequency is set at 20 kHz and the sampling frequency is fixed at 5 kHz in this simulation. The sample size N in (9) is 300, and R_0 is set to 1 Ω . The test is run for 20 times with different measured data and the NRMSE of the estimation results are depicted in Fig. 12 to illustrate the influence of the load dynamics on the estimation result.

The result in Fig. 12(a) shows that both BBO and RLS algorithm can tolerate the unknown dynamics of the load and BBO

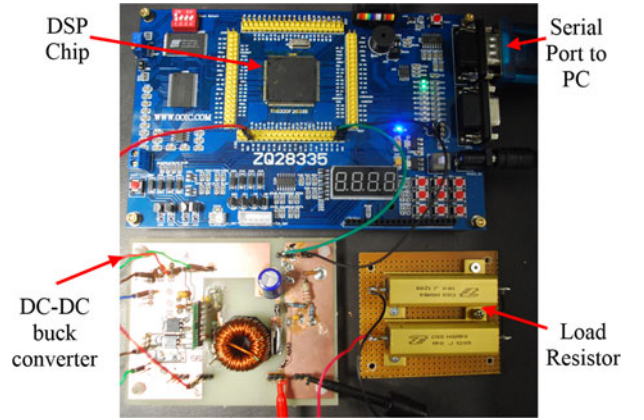


Fig. 13. Experimental setup.

is less sensitive. In Fig. 12(b), when $T_R \leq T$ the estimation error is small. When $T < T_R < 2NT$, the estimation accuracy will be affected by the load perturbation. When $T_R > 2NT$, the perturbation of the load will not affect the accuracy of the estimation. In the actual scenario, both SSR and sample size N can be adjusted to eliminate the load perturbation effect if the period of the perturbation is known.

IV. EXPERIMENTAL RESULTS

The proposed approach has been realized using a digital signal processor TMS320F28335. The experimental setup is illustrated in Fig. 13. The component and parasitic values of the capacitor and the inductor have been measured with a TTI LRC400 component meter before they are built into the buck converter. Their values are listed in Table V. The PWM switching frequency of this dc-dc buck converter is set to 20 kHz.

The input and output voltages of the converter are 10 and 3.3 V, respectively. The output is connected to a 2 Ω load resistor. Therefore, the output power in this setup is 5.4 W, and the maximum power for this converter circuit at 3.3 V output voltage is 19.8 W. The PRBS sequence is injected when the start identification command is initiated. For the system identification, 1000 samples of v_o and i_L are collected before the PRBS is disabled. The 12 bit ADC onboard of the DSP provides a resolution of less than 2 mV for the given measurement range of 0–6 V. However, the actual measurement noise of the voltage has a deviation of 7 mV. Based on the results illustrated in Fig. 7, the minimum SSR is determined as 7.

The measurement and the estimation results of the RLS method are depicted in Fig. 14. Fig. 14(a) shows the measurement waveforms of v_o and i_L . v_o is measured with the ADC port of the DSP after a voltage divider and i_L is measured with a current sensor (i.e., INA193AIDBVT) which is shunt to a 0.02 Ω sensing resistor. The error of the current sensor is 3% in the worst scenario. In Fig. 14(b), the transfer function coefficients estimation results show some couplings due to the measurement noise. Consequently, the parameter estimation result in Fig. 14(c) is unstable. It is observed that there is no estimation result for all the four parameters between 0.2 and 0.23 s. This is due to the

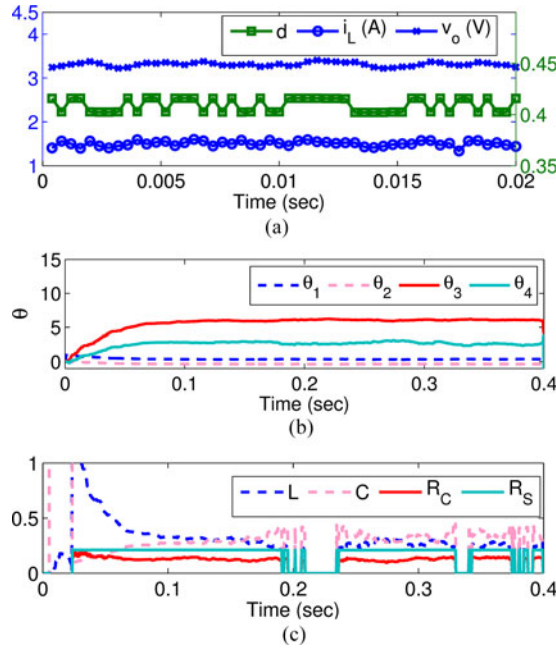


Fig. 14. (a) v_o, i_L , and d waveforms. (b) Estimation result of the discrete transfer function parameter θ . (c) Estimation result for L, C, R_C, R_S after parameter mapping.

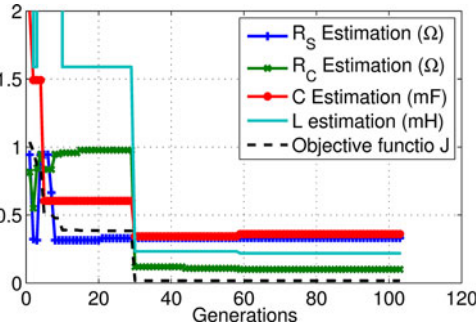


Fig. 15. BBO evolution process with the experimental data.

fact that the parameter mapping equations do not always have a feasible solution [23]. When this happens, each estimated variable will be assigned as zero. Among the available results where the mapping solutions are achievable, the estimated capacitance and inductance are also varying from time to time. In this case, it is not meaningful to evaluate the estimation accuracy.

The BBO approach is next evaluated. The data sequence of v_o, i_L , and d shown in Fig 14(a) is used for the BBO estimation. The evolution process is illustrated in Fig. 15.

Fig. 15 shows that the BBO's estimation converges at 21st generation and the cost function has decreased to 0.016 which is expected according to (10) at a noise deviation of 7 mV. The BBO parameter estimation is repeated with different data sequences for 100 times for SSR varied from 7 to 10. The averaged results are summarized in Table VI. Using the measured values from LRC as reference, the NRMSE calculated exclude the R_s estimation due to the unmeasurable series losses.

TABLE VI
MODEL PARAMETERS FOR BOOST CONVERTER EXAMPLE

	C (mF)	L (mH)	R_c (Ω)	R_s (Ω)	NRMSE%
LRC meter	0.335	0.223	0.124	0.163 (R_L)	—
SSR = 7	0.3972	0.2108	0.1105	0.3501	10.75
SSR = 8	0.3399	0.2358	0.1116	0.3308	8.27
SSR = 9	0.3523	0.2398	0.1235	0.3364	6.48
SSR = 10	0.3461	0.2443	0.1213	0.3396	9.55

The estimation accuracy achieved in the experiment is in good agreement with the simulation result in Fig. 7 at the specified noise level. As shown in Fig. 7, better performance is expected for lower noise level. It is also observed from Table V that the estimation results of the parameters are consistent regardless of the SSR settings. It should be noted that the averaged circuit model becomes inaccurate with increased sampling time as discussed in Section II. Therefore, the performance is not improved when the SSR is increased from 9 to 10. The tradeoff between the model accuracy and noise rejection should be considered in the actual applications. For example, the best performance of the proposed method is achieved with the sampling frequency of 2.22 kHz for the noise level at 7 mV deviation.

V. CONCLUSION AND FUTURE WORK

In this paper, a new method to realize the online full parameter monitoring of a dc–dc converter has been presented. The parameter estimation algorithm has been developed using BBO multivariable optimization approach. Compared with the traditional RLS estimation method, the BBO can provide a stable solution under noisy deviation of 7 mV in the experiment. Although the computational load for the BBO algorithm is generally higher than the RLS algorithm as shown in Table III, the approach is developed for online monitoring but not real-time applications. For example, the parameter identification can be implemented as a background task for a satellite power supply system to monitor the health status of the converter. Through the experiments, it is found that the estimation accuracy is improved by 4% by increasing the SSR from 7 to 9 during identification. The corresponding sampling frequency is only 2–3 kHz for a dc–dc buck converter with the MOSFETs switching at 20 kHz. The low sampling frequency property made the proposed method suitable for more applications as compared with the previous component online monitoring algorithms. The present method assumes a fixed input voltage connected converter. For the future work, this method can be generalized for converters connected to the variable input sources such as solar cells by measuring the input voltage as a second perturbation source.

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Bing-Xuan Li was born in Xi'an, China, on November 14, 1988. She received the B.Eng. degree from Nanyang Technological University, Singapore, in 2011, where she is currently working toward the Ph.D. degree.

Her current research interests include power converter fault diagnostics and control, and the modelling of failures in the power systems.



Kay-Soon Low (M'88–SM'00) received the B.Eng. degree in electrical engineering from the National University of Singapore, Singapore, and the Ph.D. degree in electrical engineering from the University of New South Wales, Sydney, Australia.

He has worked in the academia as well as in the industry. He joined the School of Electrical and Electronic Engineering, Nanyang Technological University in 1994 first as a Lecturer and subsequently became an Associate Professor. He has successfully supervised 47 graduate theses and delivered 45 funded projects. He has served as Consultants to many companies and has 19 patents on nonlinear circuits, ultrawideband radio systems, and imaging sensor. His funded projects are in the field of wireless sensor network, power electronics, solar energy, and satellite system. He has been the Centre Director of Satellite Research Centre (SaRC), Nanyang Technological University since April 2009. The Centre has successfully developed four satellites (X-SAT, VELOX-P, VELOX-PII, and VELOX-PIII) and they have been launched between April 2011 and June 2014. In orbit experiments such as fault tolerant power management system, peak power tracker for solar energy, model predictive control of satellite attitude have been successfully demonstrated.

Dr. Low received the IEEE Industrial Electronics and Applications 2014 Best Conference Paper Award, the Singapore Public Administration Medal (Bronze) in 2014, and Defence Technology Prize (Research & Development category) in 2011.